



22MA201

ENGINEERING MATHEMATICS II

UNIT 1 & LP1 –FORMATION OF DIFFERENTIAL EQUATIONS

1.Ordinary and partial differential equations

As stated earlier, an equation involving one dependent variable and its derivatives w.r.t one or more independent variables is called a differential equation.

If a differential equation contains only one independent variable (and hence ordinary derivatives), it is called an ordinary differential equation. The acronym ODE is commonly used.

Ex: With 'y' as dependent variable and 'x' as the independent variable, two ODEs are given below

i) $\frac{dy}{dx} - 5x = \sin x$

ii) $\left(\frac{dy}{dx}\right)^3 + 2\frac{d^2y}{dx^2} - e^x = 5$

If a differential equation contains two or more independent variables (and hence partial derivatives), it is called a partial differential equation.

Ex: With 'u' as dependent variable and x, y & z as independent variables, an example of partial differential equation is

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 1$$

We know that derivative is a rate of change.

Hence description of changes in state or motion involve differential equations.

1.1.Order and Degree of a differential equation

Let a differential equation be expressed as a polynomial equation in the derivatives occurring in it.

The exponent of each of the derivatives should be minimum. Exponent of the variables need not be an integer.

Under such conditions,

- i) the order of the differential equation is the order of the highest derivative in it.
- ii) the degree of the differential equation is the largest exponent of the highest order derivative.

Note: If the differential equation cannot be expressed as a polynomial of derivatives, its 'degree' is not defined.

These are explained by the following examples.

Ex 1:

$$3 \left(\frac{d^3 y}{dx^3} \right)^2 + \left(\frac{d^2 y}{dx^2} \right) - 2 \left(\frac{dy}{dx} \right)^3 = 5$$

It is polynomial in $\frac{dy}{dx}$, $\frac{d^2 y}{dx^2}$ and $\frac{d^3 y}{dx^3}$.

$\frac{d^3 y}{dx^3}$ is the highest order derivative.

Hence the order of the differential equation is 3.

Now, the exponent of $\frac{d^3 y}{dx^3}$ is 2, that of $\frac{d^2 y}{dx^2}$ is 1 and that of $\frac{dy}{dx}$ is 3.

For finding the degree, we have to consider the exponent of highest order derivative which is of third

order ($\frac{d^3 y}{dx^3}$) in the given equation. Its exponent (or power) is 2.

Therefore, the order of the given differential equation is 3 and its degree is 2. Hence the degree is 2.

The general form of an ordinary differential equation of nth order is

$$F \left(x, y, \frac{dy}{dx}, \dots, \frac{d^n y}{dx^n} \right) = 0$$

or $F(x, y, y_1, y_2, \dots, y_n) = 0$



or $F(x, y, y', y'', \dots, y(n)) = 0$

Note that, it is not mandatory that derivatives of all orders need to be present in a differential equation.

<https://www.simply.science/index.php/math/calculus/differential-equations/formation-of-differential-equations>

1.2. Formation of ordinary differential equation:

Consider the equation $f(x, y, c_1) = 0$ -----(1) where c_1 is the arbitrary constant. We form the differential equation from this equation. For this, differentiate equation (1) with respect to the independent variable occurring in the equation.

Eliminate the arbitrary constant c from (1) and its derivative. Then we get the required differential equation.

Suppose we have $f(x, y, c_1, c_2) = 0$. Here we have two arbitrary constants c_1 and c_2 . So, find the first two successive derivatives. Eliminate c_1 and c_2 from the given function and the successive derivatives. We get the required differential equation.

If an equation in independent and dependent variables involving some arbitrary constants is given, then a differential equation is obtained as follows:

(i) Differentiate the given equation w.r.t. the independent variable (say x) as many times as the number of arbitrary constants in it.

(ii) Eliminate the arbitrary constants.

(iii) The elimination is the required differential equation.

For eg:- Form the differential equation, if $y = e^{2x} + x + C$ is the general solution.

$$y = e^{2x} + x + C$$

Since y depends on x , so differentiate above equation w.r.t. x and we get

$$y' = 2e^{2x} + 1$$

Hence, the differential equation formed from y is $y' = 2e^{2x} + 1$.

1.3. Formation of differential equation whose general solution is of the form $F(x, y, a) = 0$

Problem 1:



Let us consider the equation

$$x^2 + y^2 = r^2$$

Solution:

By giving different values to r , we get different members of the family

e.g. $x^2 + y^2 = 1$, $x^2 + y^2 = 4$, $x^2 + y^2 = 9$, etc.

Thus, the above equation represents a differential equation for a family of concentric circles centred at the origin and having different radii.

We are interested in finding the differential equation for these concentric circles.

Since, r is the only arbitrary constant in the general solution, so we differentiate it once w.r.t. x

$$2x + 2y \frac{dy}{dx} = 0$$

$$\text{or, } x + y \frac{dy}{dx} = 0$$

which represents a differential equation for the family of concentric circles given by the equation $x^2 + y^2 = r^2$.

Problem 2:

Find the differential equation of the family of circles $x^2 + y^2 = 2ax$, where a is a parameter.

Solution:

Given, $x^2 + y^2 = 2ax$ (1) By differentiating both the sides of (1) with respect to x , we get,

$$2x + 2y \frac{dy}{dx} = 2a \text{(ii)}$$

Now, eliminating a from (i) and (ii) we get,

$$x^2 + y^2 = x(2x + 2y \frac{dy}{dx}) \text{ or, } 2xy \frac{dy}{dx} = y^2 - x^2$$

Which is the required differential equation of the family of circles (1).

1.4. Formation of differential equation whose general solution is of the form $F(x, y, a, b) = 0$ **Problem 3:**

Form the differential equation representing the family of curves $y = a \sin(x+b)$, where a, b are arbitrary constants.

Solution: - We have

$$y = a \sin(x+b) \text{ ... (1)}$$

Differentiating both sides of equation (1) with respect to x , successively we get

$$\frac{dy}{dx} = a \cos(x+b) \text{ ... (2)}$$

$$d^2y/dx^2 = -a \sin(x+b) \dots (3)$$

Eliminating a and b from equations (1), (2) and (3), we get

$$d^2y/dx^2 + y = 0 \dots (4)$$

which is free from the arbitrary constants a and b and hence this is the required differential equation.

Problem 4:

Find the differential equation of the family of curves $y = a/x + b$ where a and b are arbitrary constants.

Solution:

Given $y = \frac{a}{x} + b$

Differentiating w.r.t x, we get

$$\frac{dy}{dx} = \frac{-a}{x^2}$$
$$x^2 \frac{dy}{dx} = -a$$

Again differentiating w.r.t x we get

$$x^2 \frac{d^2y}{dx^2} + 2x \frac{dy}{dx} = 0$$
$$\Rightarrow x \frac{d^2y}{dx^2} + 2 \frac{dy}{dx} = 0$$

which is the required differential equation

Problem 5:

Find the differential equation corresponding to $y = ae^{4x} + be^{-x}$ where a, b are arbitrary constants,

Solution:

Given $y = ae^{4x} + be^{-x}$. (1)

Here a and b are arbitrary constants

From (1), $\frac{dy}{dx} = 4ae^{4x} - be^{-x}$ (2)

and $\frac{d^2y}{dx^2} = 16ae^{4x} + be^{-x}$ (3)

(1)+(2) $\Rightarrow y + \frac{dy}{dx} = 5ae^{4x}$ (4)

(2)+(3) $\Rightarrow \frac{dy}{dx} + \frac{d^2y}{dx^2} = 20ae^{4x}$
 $= 4(5ae^{4x})$
 $= 4\left(y + \frac{dy}{dx}\right)$

$\frac{dy}{dx} + \frac{d^2y}{dx^2} = 4y + 4\frac{dy}{dx}$
 $\Rightarrow \frac{d^2y}{dx^2} - 3\frac{dy}{dx} - 4y = 0,$

which is the required differential equation.

Note

The order of the differential equation to be formed is equal to the number of arbitrary constants present in the equation of the family of curves.

Problem 6:

Find the differential equation of the family of straight lines $y=mx+c$ when

- (i) m is the arbitrary constant
- (ii) c is the arbitrary constant
- (iii) m and c both are arbitrary constants.

Solution:

- (i) m is an arbitrary constant

$$y = mx + c \quad \dots(1)$$

Differentiating w.r. to x , we get $\frac{dy}{dx} = m \quad \dots(2)$

Now we eliminate m from (1) and (2)

For this substitute (2) in (1)

$$y = x \frac{dy}{dx} + c$$

$$x \frac{dy}{dx} - y + c = 0 \quad \text{which is the required differential equation of first order.}$$

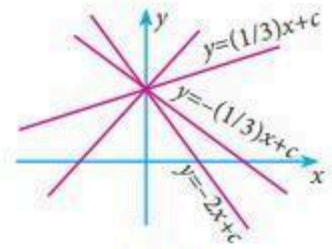


Fig. 4.1

- (ii) c is an arbitrary constant

Differentiating (1), we get $\frac{dy}{dx} = m$

Here c is eliminated from the given equation

$$\therefore \frac{dy}{dx} = m \quad \text{is the required differential equation.}$$

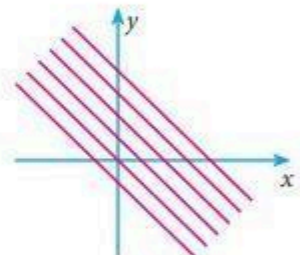


Fig. 4.2

- (iii) both m and c are arbitrary constants

Since m and c are two arbitrary constants differentiating (1) twice we get

$$\frac{dy}{dx} = m$$

$$\frac{d^2y}{dx^2} = 0$$

Here m and c are eliminated from the given equation.

$$\frac{d^2y}{dx^2} = 0 \quad \text{which is the required differential equation.}$$

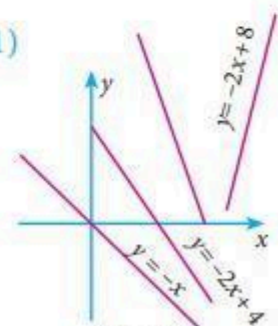


Fig. 4.3

Problem 7:

Find the differential equation of the family of curves $y = e^x (a \cos x + b \sin x)$ where a and b are arbitrary constants.

Solution :

$$y = e^x (a \cos x + b \sin x) \quad (1)$$

Differentiating (1) w.r.t x , we get

$$\begin{aligned} \frac{dy}{dx} &= e^x (a \cos x + b \sin x) + e^x (-a \sin x + b \cos x) \\ &= y + e^x (-a \sin x + b \cos x) \quad (\text{from (1)}) \end{aligned}$$

$$\Rightarrow \frac{dy}{dx} - y = e^x (-a \sin x + b \cos x) \quad (2)$$

Again differentiating, we get

$$\frac{d^2y}{dx^2} - \frac{dy}{dx} = e^x (-a \sin x + b \cos x) + e^x (-a \cos x - b \sin x)$$

$$\frac{d^2y}{dx^2} - \frac{dy}{dx} = e^x (-a \sin x + b \cos x) - e^x (a \cos x + b \sin x)$$

$$\frac{d^2y}{dx^2} - \frac{dy}{dx} = \left(\frac{dy}{dx} - y \right) - y \quad (\text{from (1) and (2)})$$

$$\Rightarrow \frac{d^2y}{dx^2} - 2 \frac{dy}{dx} + 2y = 0, \text{ which is the required differential equation.}$$

Discussion Questions:

1. Is the order and degree of a differential equation the same?
2. What is the derivative of a function?
3. How many times we have to differentiate the given equation to form a differential equation?